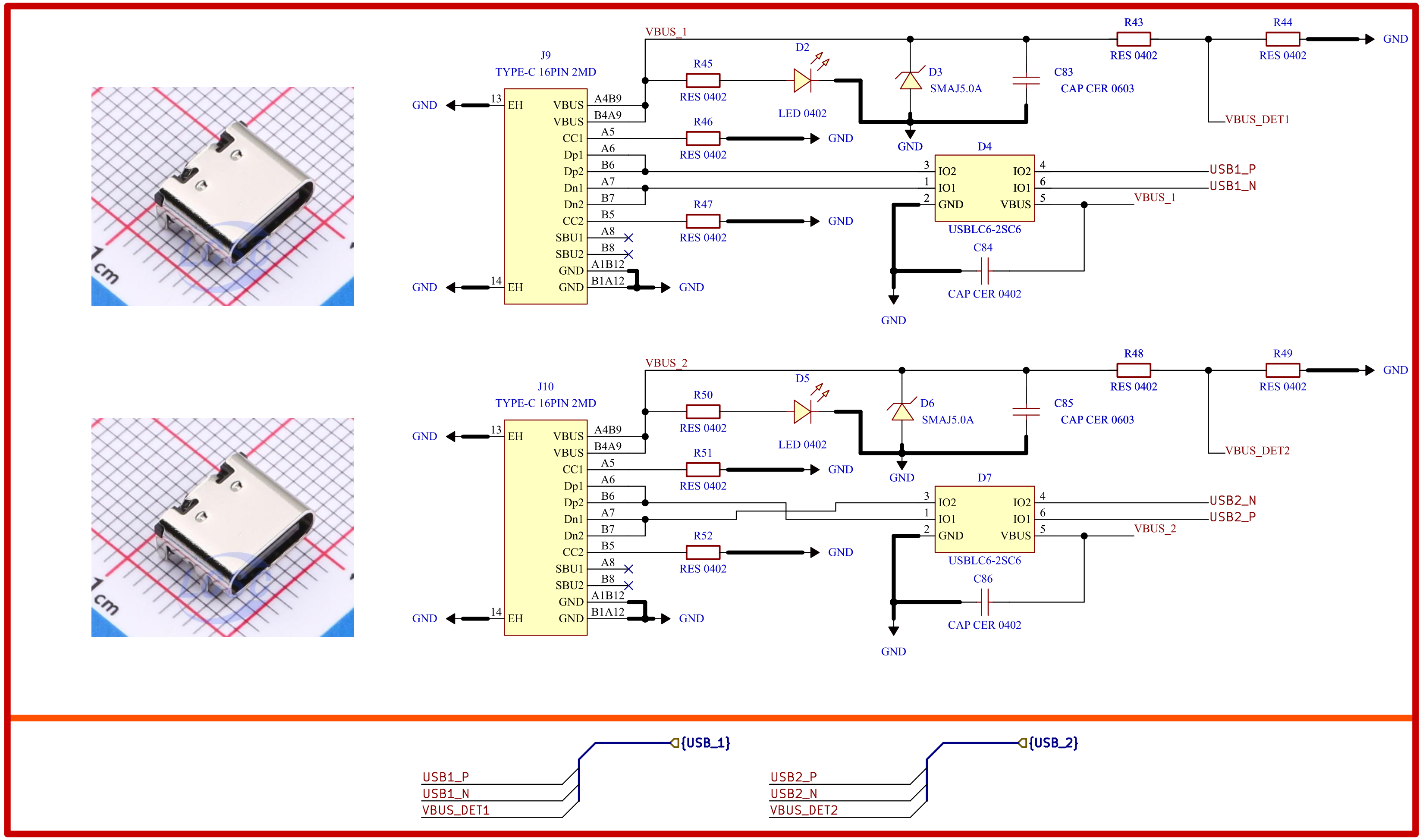
[14] Interface - USB 2.0



DESIGN NOTE:
[1] VBUS NOT connected to +5V/+3V3. Board is self-powered(LiPo). Sense + LED only. No backfeed to host.

[2] LED direct from VBUS @ 1mA. USB2.0 suspend limit 2.5mA. LED 1mA + divider 33uA = 1.03mA (41%). Do NOT raise to 5mA.

[3] VBUS_DET required - G474 has no dedicated VBUS pin. Self-powered device must not pull up D+ when VBUS=0. FW controls DPPU bit in USB_BCDR. 68k/100k: VBUS 4.40V -> 2.60V > VIH 2.31V. Debounce 100ms.

[4] D+/D- direct to PA12/PA11. No 22R series. No external 1k5 pull-up (internal RPUI 900-1500R).

[5] USB clock: PLL-Q 48MHz from HSE 8MHz. ST DFU bootloader uses HSI48+CRS - flashable without crystal.

[6] Both USB ports fully isolated - no shared nets.

[7] GROUND LOOP WARNING: USB from PC ties host GND to power GND (4.6A peak @20kHz). Use isolator (ADuM3160) or isolated USB-UART when debugging with motors running.

Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)		Variant: DRAFT	Git Hash: 7187a9a
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: Interface - USB 2.0.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
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